ESPRIT SCHOOL OF BUSINESS · INTEGRATED PROJECT 2026

VIGIE

Customer churn analysis and prediction

See churn coming. Act before the account closes.

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INTRODUCTION

The question: understand, measure and anticipate churn

- **Anonymised banking snapshot**: 483,536 rows · 363,024 customers · 410,309 accounts
- **Churn = account closed**; a customer churns when all accounts are closed
- **Five phases**: data → warehouse → reporting → prediction → application
- **One rule**: every figure is verified — checks, reference values, reproducible models
- **Privacy by design**: no data in the repository, aggregates only in the app

36.1%
churn rate per account

36.8%
churn rate per customer

51%
of closures in 5 mass weeks

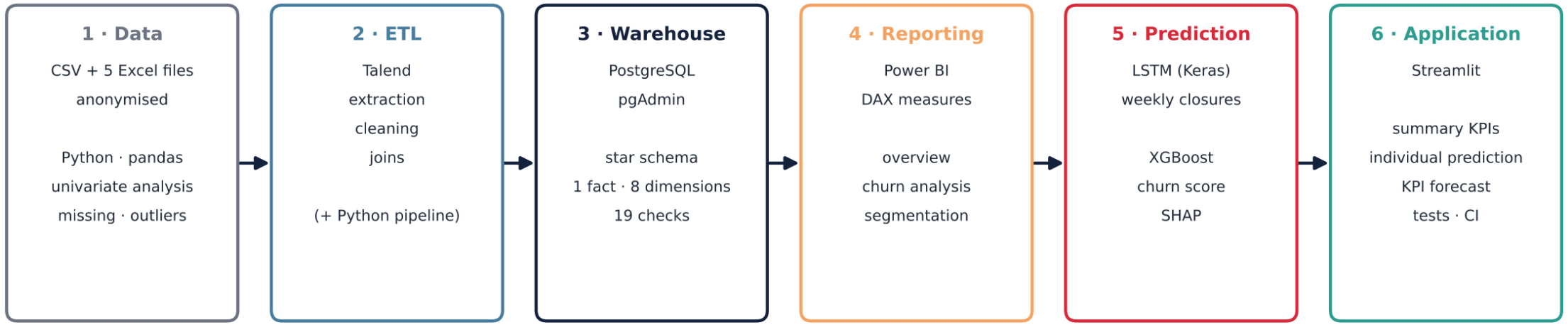
0.751
ROC-AUC of the churn model

19 / 19
warehouse checks passed

26
automated tests

Architecture: one chain, one code base

Architecture of the VIGIE solution

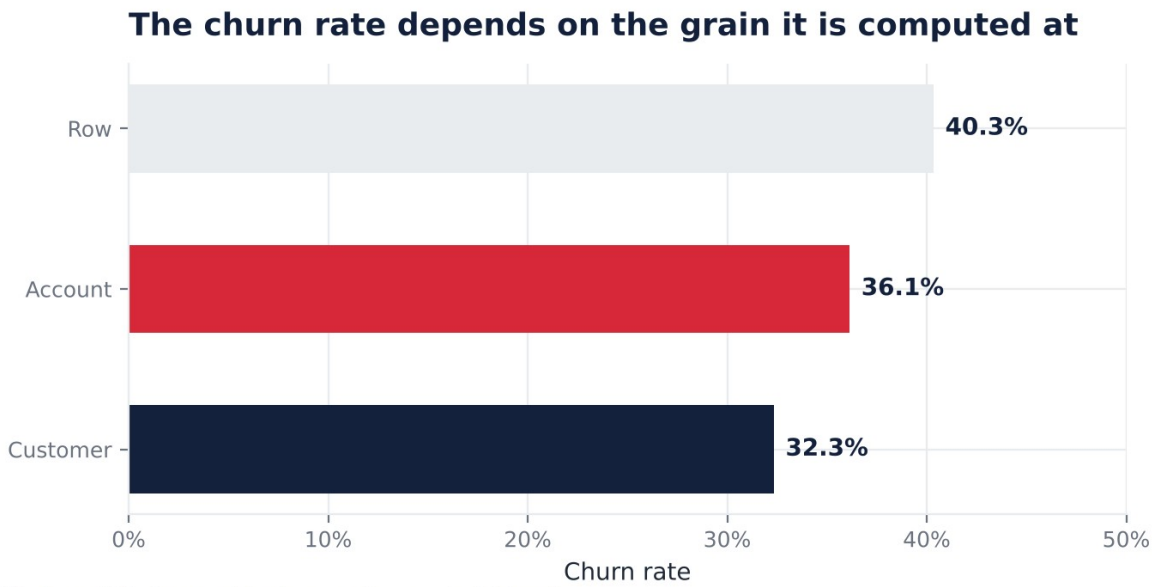


Cross-cutting foundation

Git / GitHub · shared vigie/ library (one code base for every figure) · no data in the repository · aggregates and fictitious profiles only in the application

VIGIE · Group BI 5

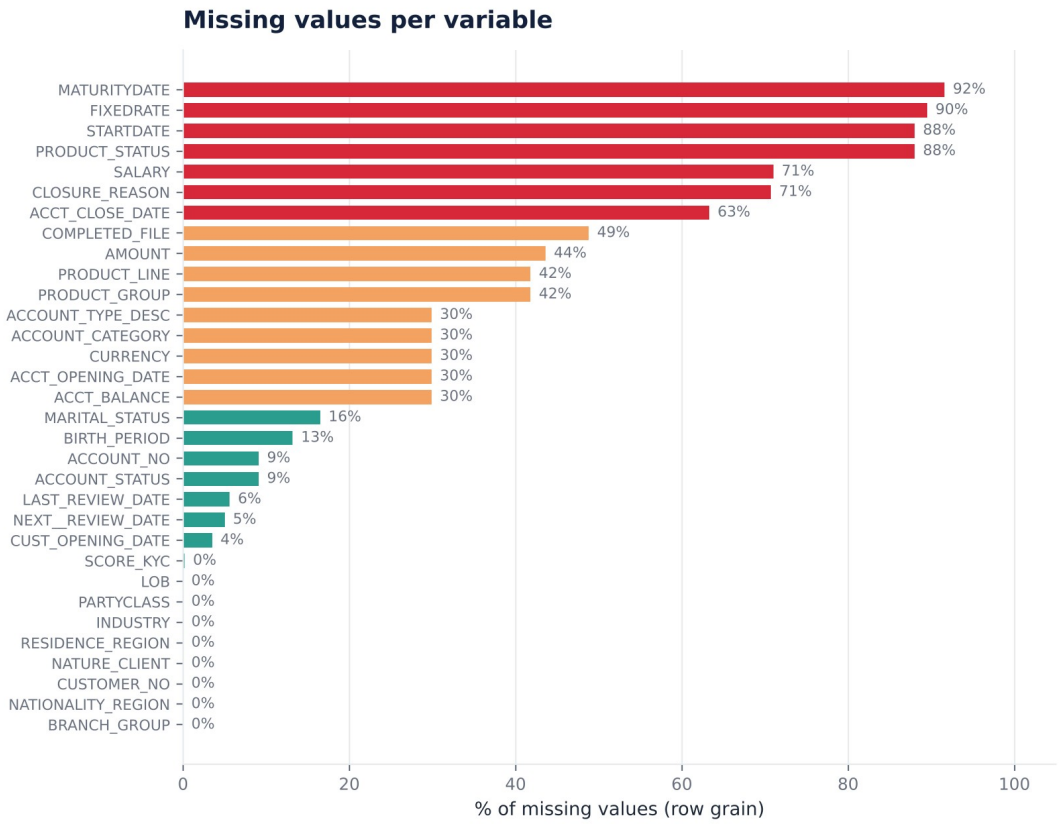
Phase 1 · A rate is meaningless without its grain



VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

- **Volumes match the supervisors' document**: to the unit
- **One row = customer × account × product**: 9% of rows have no account
- **No customer is both active and closed**: customer churn is unambiguous
- **Moderate imbalance (36.8%)**: no rebalancing needed
- **Anomaly**: 20,166 accounts closed before they were opened — flagged

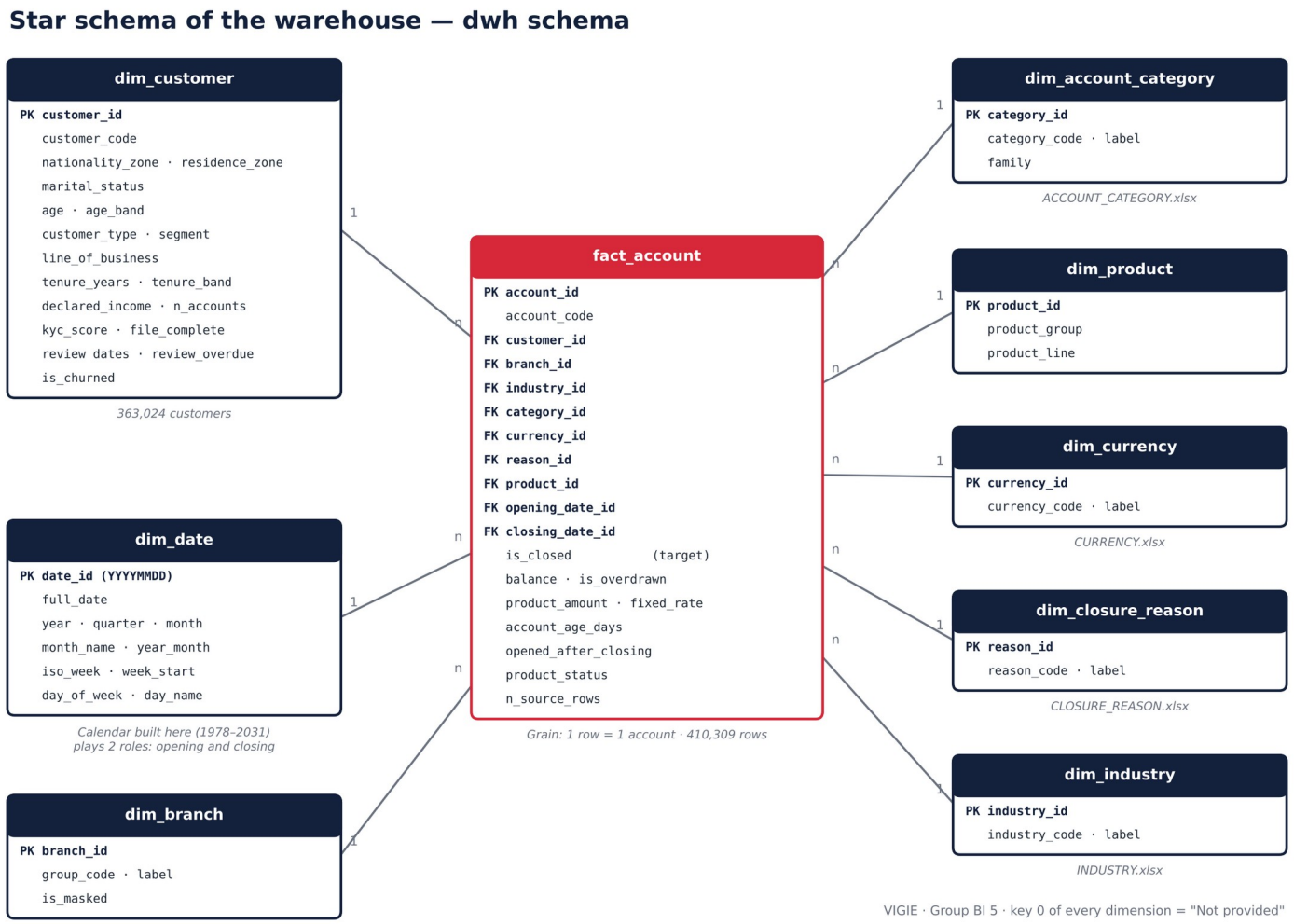
Phase 1 · Some blanks give the answer away



VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

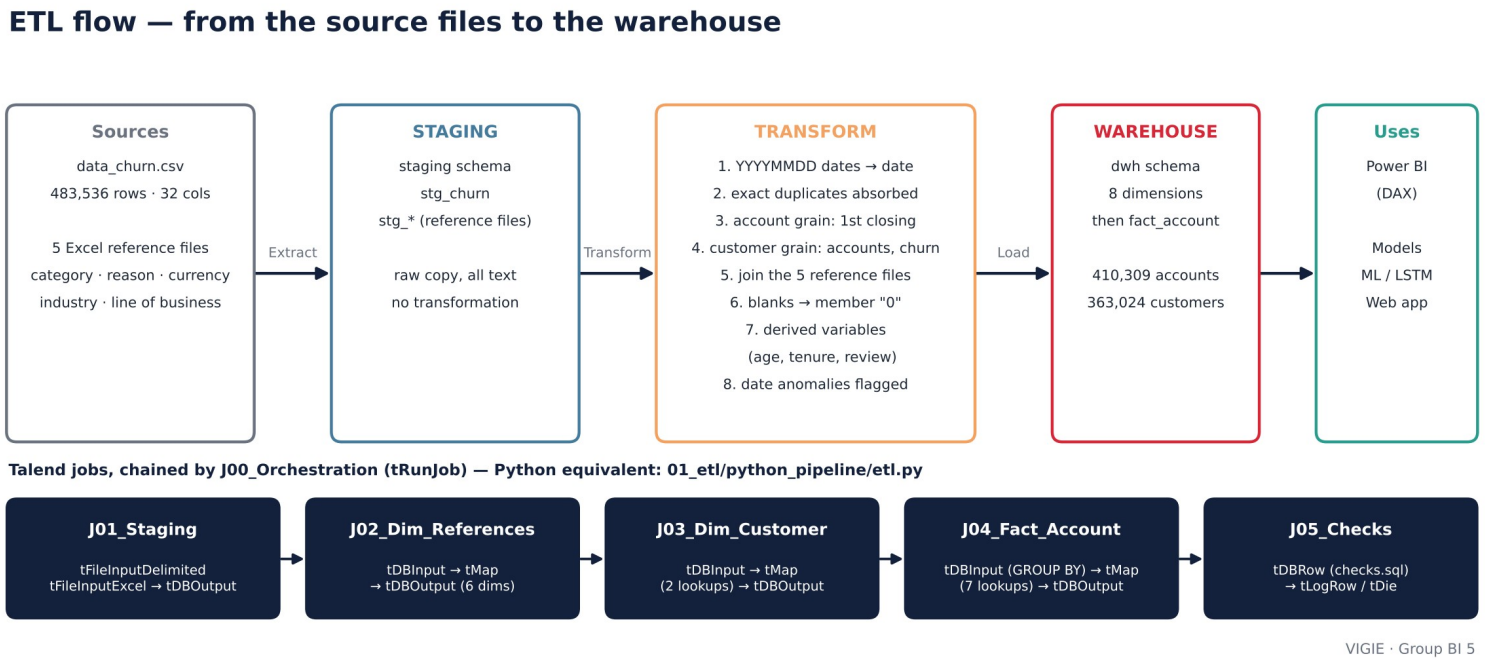
- **Structural, business, declarative blanks**: typed one by one
- **Tied to the closure**: blank for 67.8% of closed accounts, 0% of active ones → target leak
- **Outliers (IQR)**: 32% of balances — loan accounts in debit, not errors
- **No row deleted**: flags and medians instead
- **Bivariate (bonus)**: incomplete file → 49.6% churn vs 22.5%

Phase 2 · Star schema at account grain



- **fact_account**: 410,309 accounts
- **8 dimensions**: customer keeps the 43,984 customers with no account
- **Key 0 = Not provided**: no null foreign key
- **Calendar in SQL**: two roles
- **Bonus**: SCD type 2 on the customer

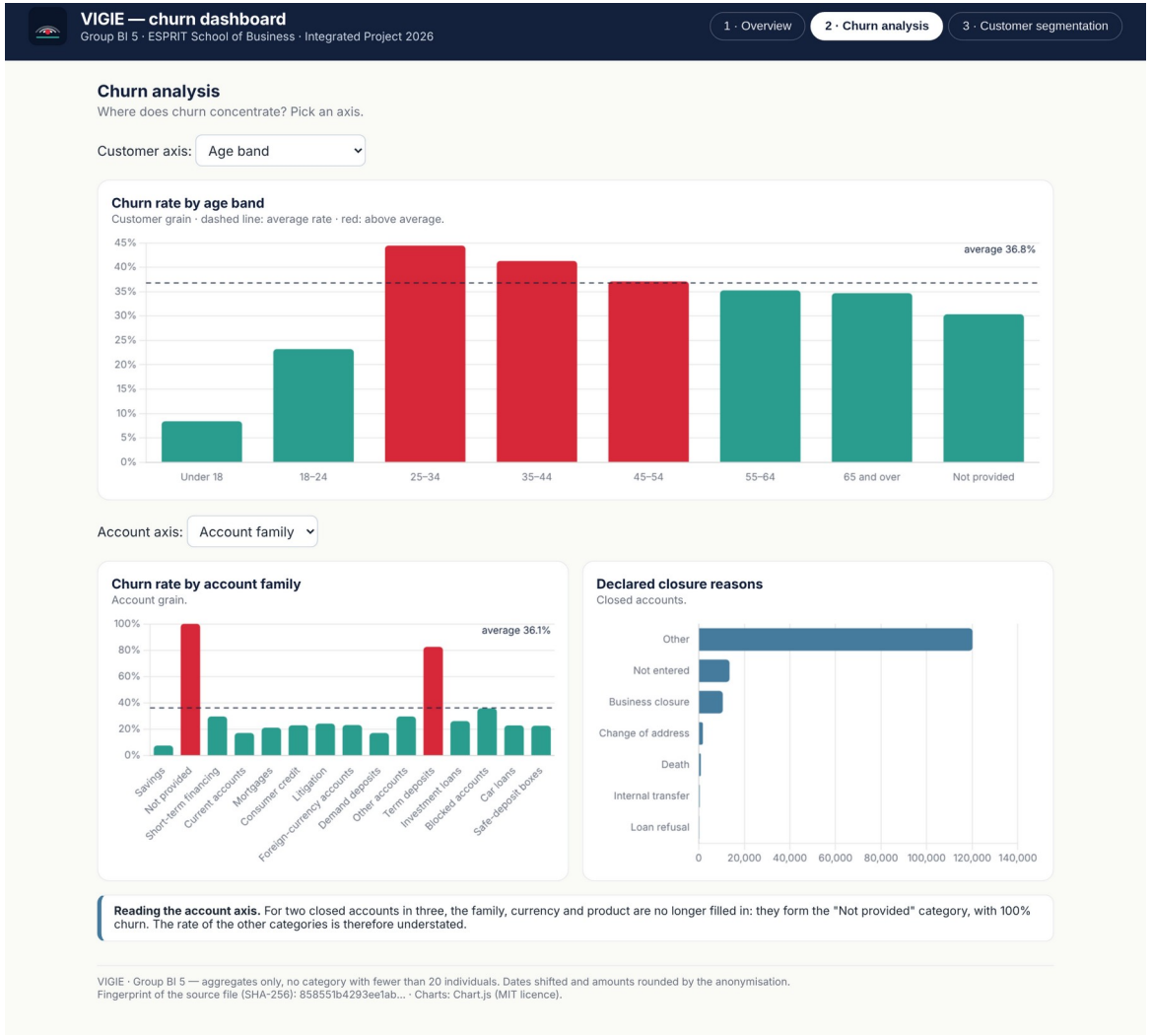
Phase 2 · ETL and 19 checks, 19 OK



- **Volumes**: 483,536 rows · 410,309 accounts · 363,024 customers — identical on both sides
- **Indicators**: 148,215 closures · churn 36.12% · first closing date kept · balances reconciled
- **Integrity**: no duplicate, no orphan, every date in the calendar, anomalies flagged

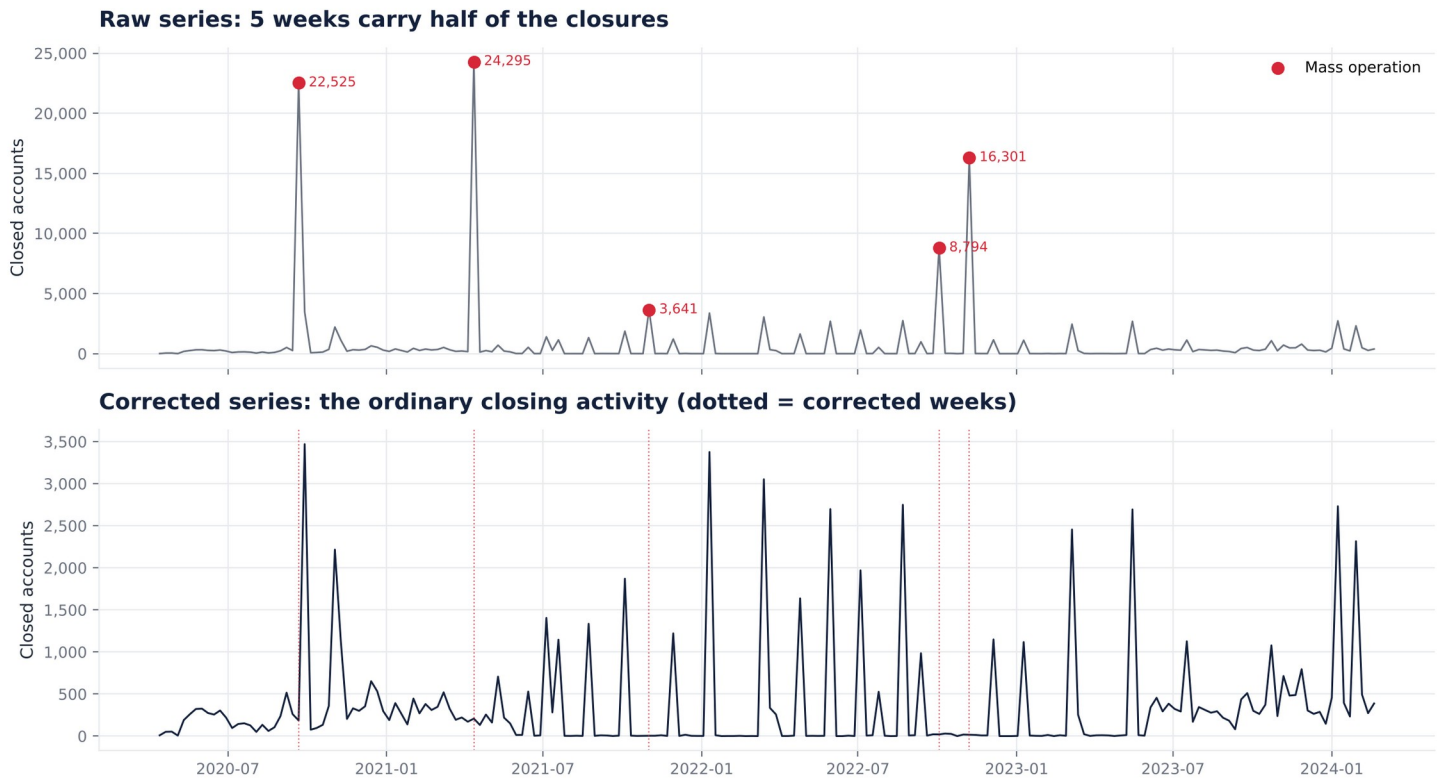
REPORTING

Phase 3 · Reporting: 3 pages, 30 documented DAX measures



- **Pages**: overview · churn analysis · segmentation
- **Account Churn Rate = 36.12%**: Customer Churn Rate = 36.78%
- **Two inactive date relationships**: USERELATIONSHIP per measure
- **"Not provided" stays visible**: 68% of closed accounts
- **Bonus**: Chart.js dashboard, offline, phone-friendly

Phase 4 · The KPI: weekly closures — and 5 mass weeks

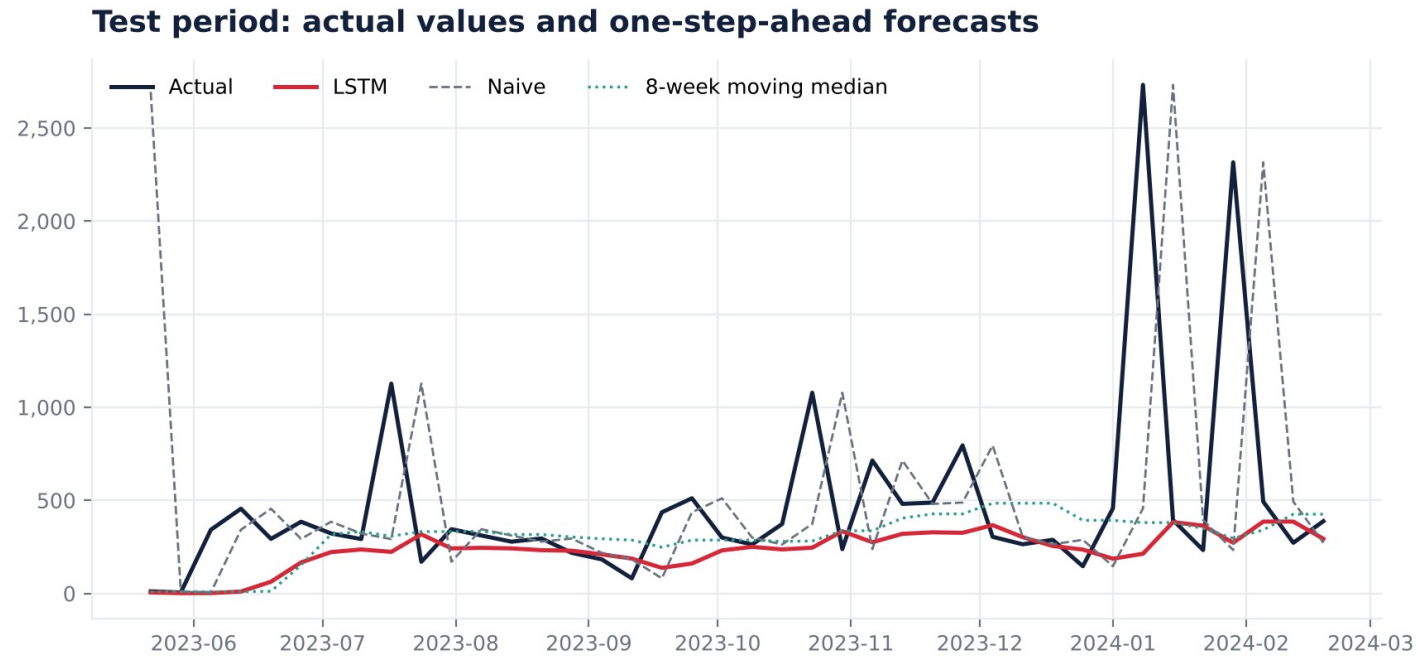


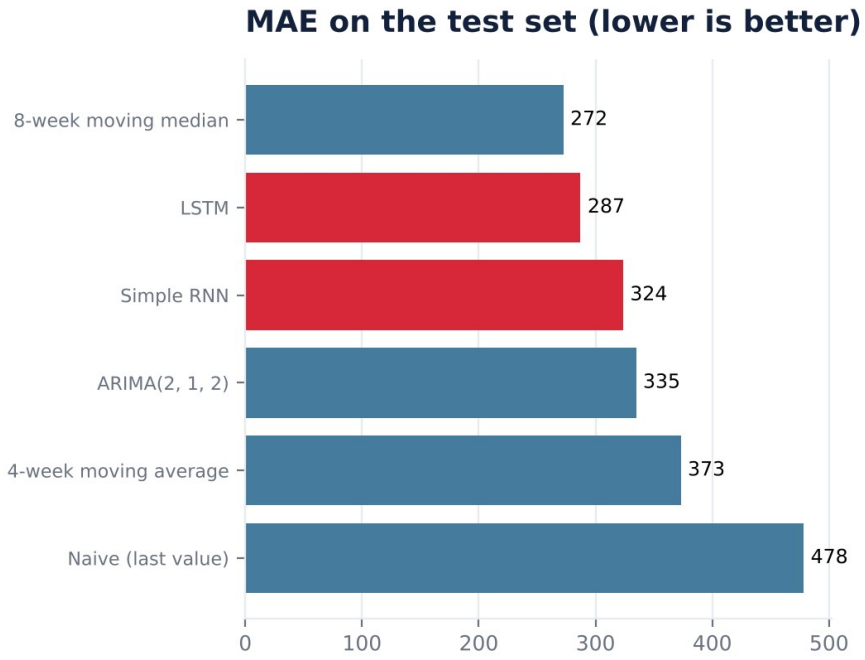
VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

- **202 weeks**: 11 zero weeks added
- **5 weeks = 51% of closures**: batch operations
- **Corrected series**: local median
- **Three regimes**: no calendar seasonality
- **8-week window**: MAE / RMSE (no MAPE)

PREDICTIVE MODELS

Phase 4 · LSTM: −40% vs naive, but a median does as well



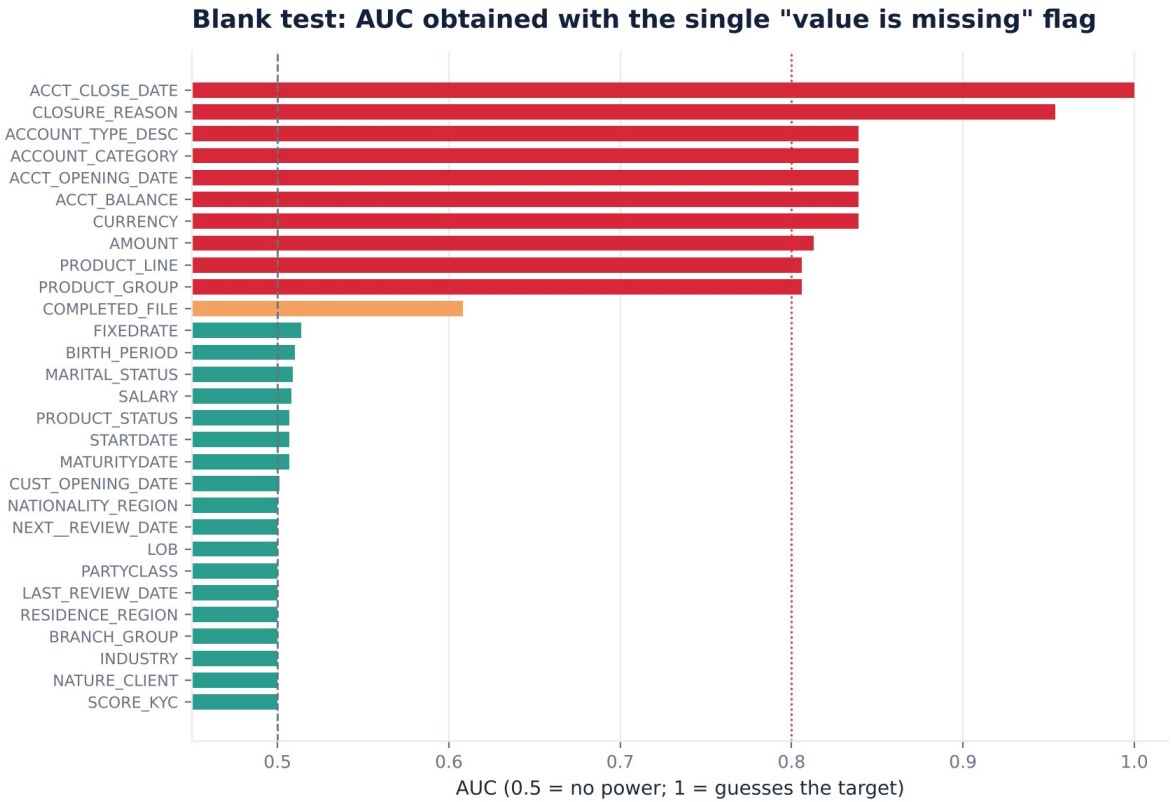


VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

- **LSTM MAE 287** · moving median 272 · RNN 324 · ARIMA 335 · naive 478 (last 40 weeks)
- **Second KPI (bonus)**: customers lost per week — the LSTM has the best MAE (125 vs 133)

PREDICTIVE MODELS

Phase 4 · Leakage check: a 0.95 AUC would be a lie



VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

0.951

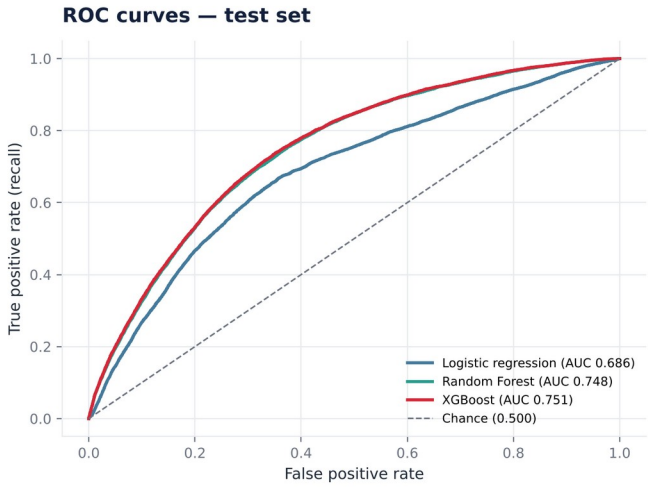
ROC-AUC with 3 leaking flags

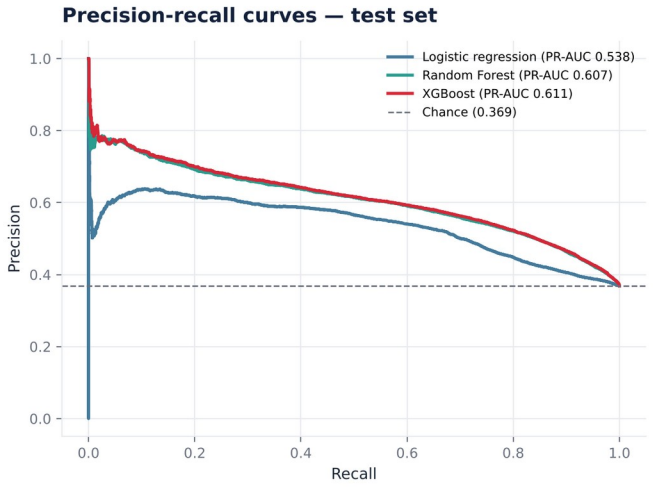
0.751

ROC-AUC without them — the model kept

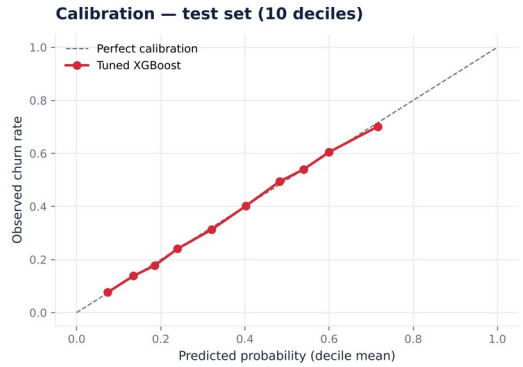
10 columns excluded: closing date and reason, balance, opening date, family, currency, amount, product. PRODUCT_STATUS excluded for its values.

Phase 4 · Churn model: XGBoost, AUC 0.751, recall 79%





VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

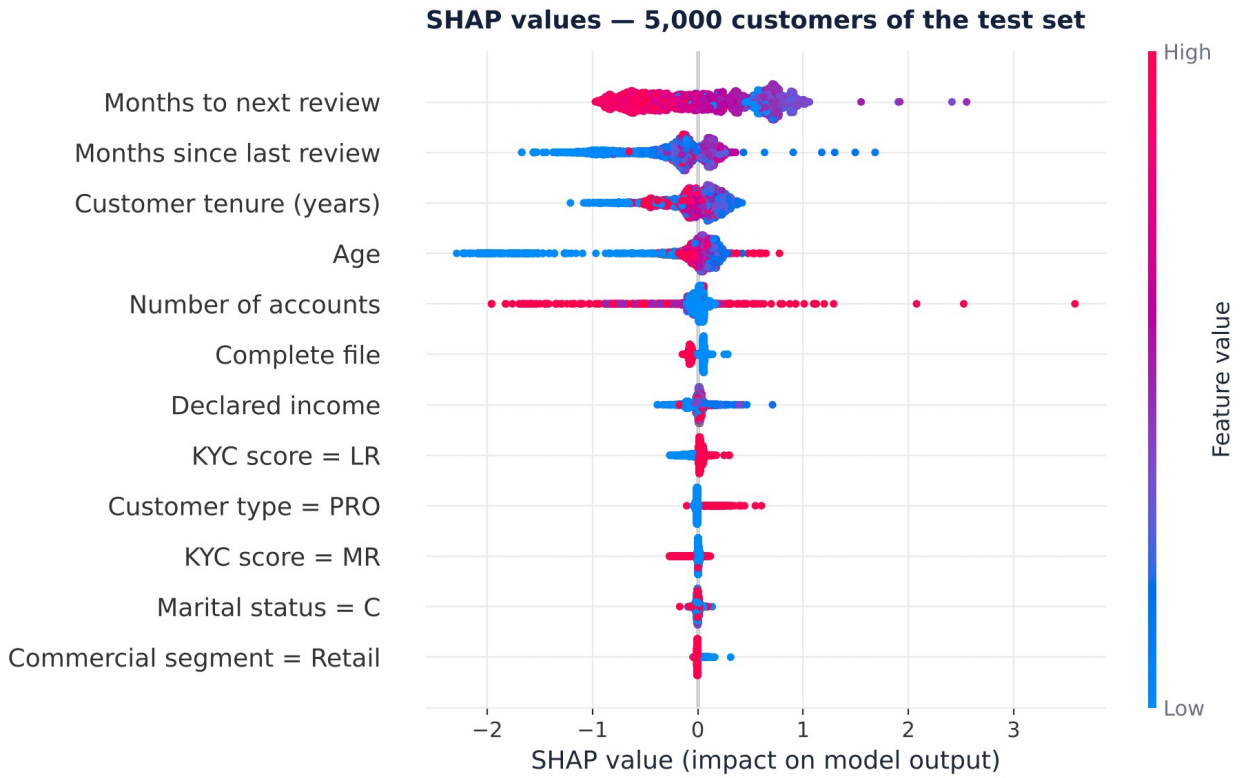




VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

- **Logistic regression**: 0.686
- **Random Forest**: 0.748
- **Tuned XGBoost**: 0.751 · PR-AUC 0.611
- **Calibrated**: within 1.5 pt per decile
- **Gains**: top 30% of scores → 50% of departures
- **Risk levels**: 15% · 38% · 61% observed departures

Phase 4 · Why customers leave

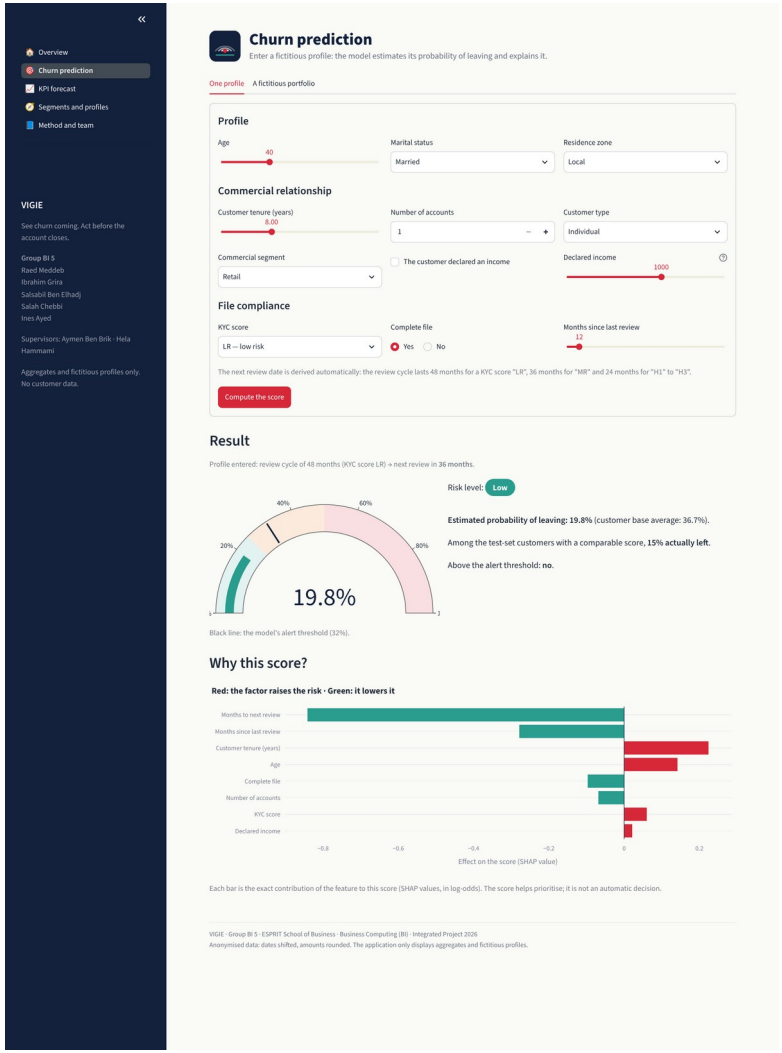


- **Overdue review**: +0.53 — the first warning sign
- **Recent review**: −0.53 — protects
- **25–44 years, 5–10 years of tenure**: the fragile zone
- **Two accounts or more**: −0.20
- **Compliance = 62% of importance**: segment < 2%
- **Honest bound**: AUC 0.751 with, 0.660 without compliance

VIGIE · Group BI 5 · Source: data_churn.csv (anonymised dataset)

APPLICATION

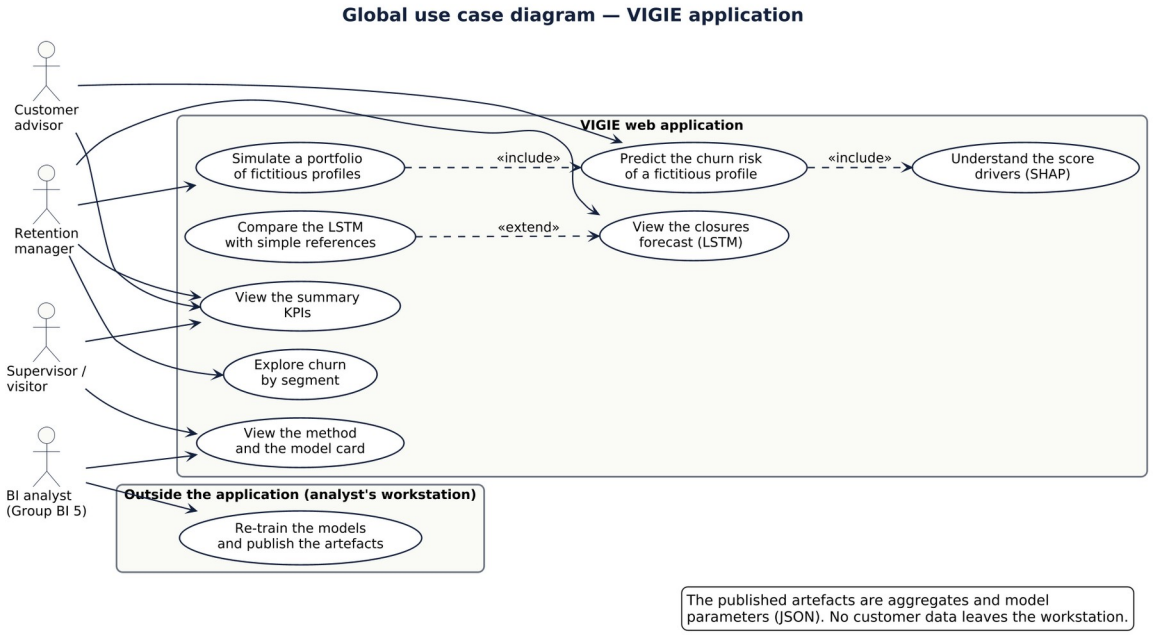
Phase 5 · The VIGIE application

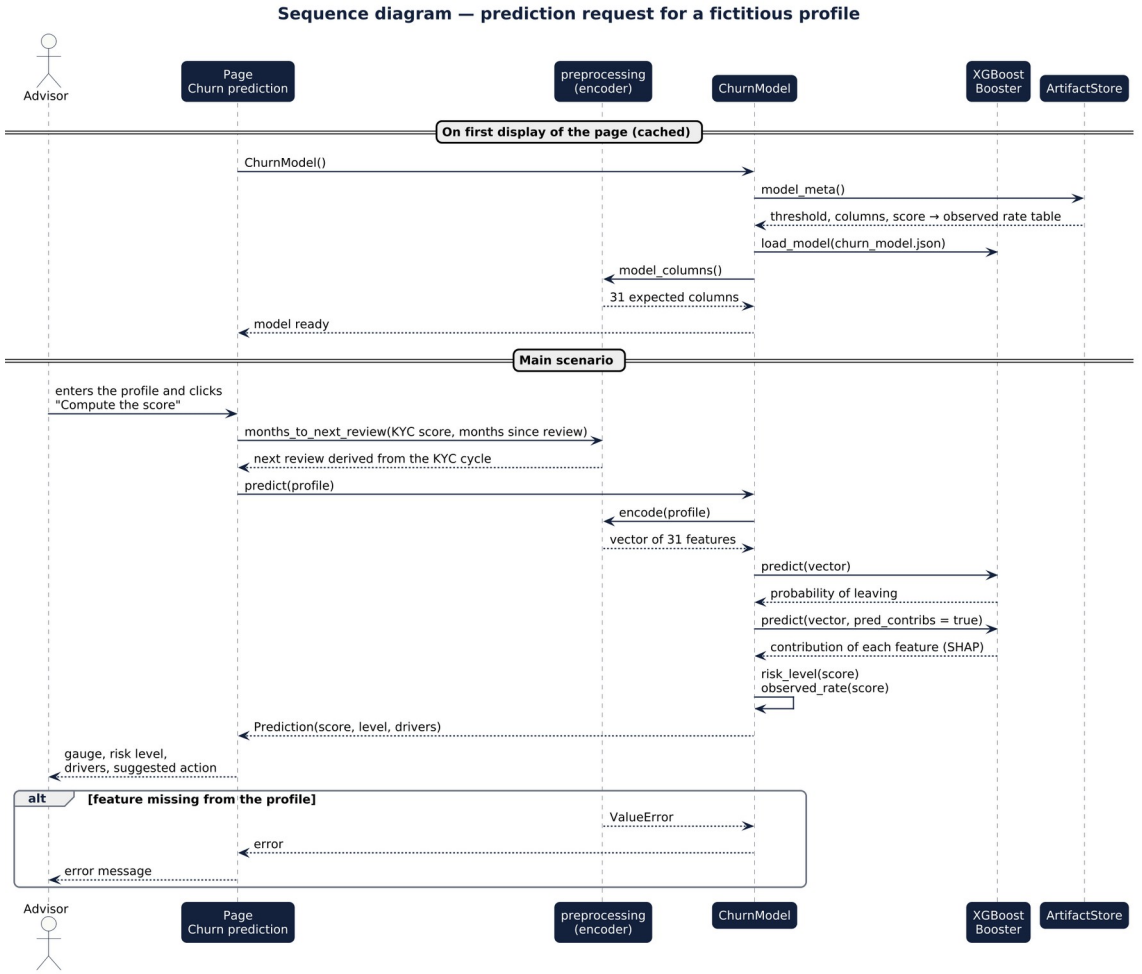


- **Overview**: 8 KPIs, churn by axis, closures and forecast
- **Prediction**: fictitious profile → score, risk level, SHAP drivers, action
- **Forecast**: LSTM 8 weeks + comparison with references
- Segments · Method and team
- **No data**: aggregates (≥ 20) and JSON trees only
- **Same encoder as training**: no gap between model and app

APPLICATION

Phase 5 · UML: use cases, sequence, classes





APPLICATION

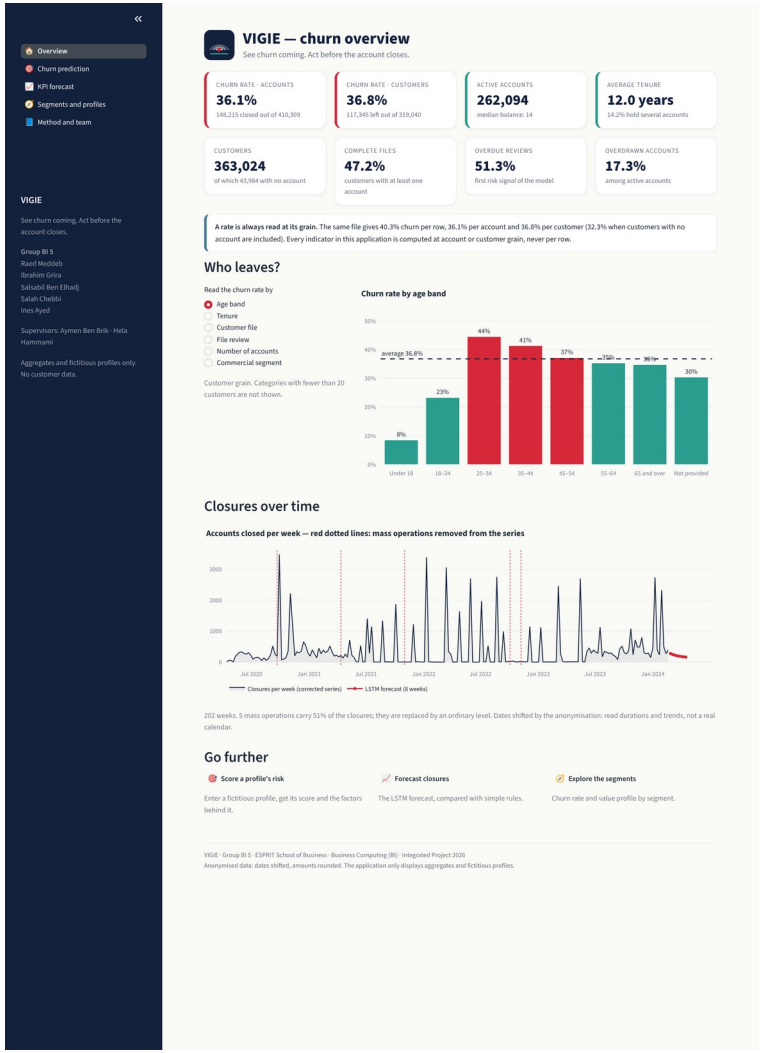
Phase 5 · Tests, CI and data protection

26	5 / 5	0	100%
automated tests (pytest)	pages tested end to end	data files in the repository	metrics reproduced by train.py

- **Encoding**: fixed columns, missing values, unknown categories, KYC review cycle
- **Model**: probabilities, exact SHAP explanation, direction of effects, fictitious portfolio
- **Application**: every page runs; the form reacts in the right direction
- **Confidentiality**: no data file, no identifier, no aggregate below 20, consistent volumes
- **Deployment**: Streamlit Community Cloud — no TensorFlow, no .pkl needed

APPLICATION

Live demo



- **1 · Overview**: KPIs and "a rate at its grain"
- **2 · Prediction**: default profile → overdue review → score rises
- **3 · Fictitious portfolio**: who to call first
- **4 · Forecast**: next 8 weeks, honest comparison
- **Backup**: recorded video

Limitations and next steps

- **A snapshot, not a film**: some features may follow the departure
- **No behavioural data**: AUC capped at 0.75
- **Batch closures**: half of the "churn"
- **202 weeks, 3 regimes**: hard to forecast

- **Separate batch closures** from real departures
- **Monthly behavioural snapshots** (balances, complaints)
- **Re-train regularly**: LSTM next to the median
- **Measure value**: campaign vs control group

Conclusion: watch the files nobody reviews

- **Complete chain**: data → warehouse → Power BI → LSTM and XGBoost → web application
- **Verified at every step**: 19 checks · reference values for every DAX measure · reproducible models · 26 tests
- **Lessons**: a rate needs its grain; blanks can fake a perfect model; a deep network does not always beat a median
- **Business message**: an overdue file review is the strongest warning sign — act on it

Thank you — questions?